

1. [15 Marks: 1, 1, 1, 1, 1]

- a) Find two numbers which add to make 22 and multiply to make 40

20 and 2

- b) Evaluate
- 12^1

$$= 12$$

- c) Evaluate
- $2^3 + 3^2$

$$8 + 9 = 17$$

- d) Evaluate
- $(-4)^2$

$$= 16$$

- e) Evaluate
- $2^0 + 2^2$

$$= 1 + 4 = 5$$

D/E

2. [12 Marks: 2, 2, 2, 2, 2, 2]

Simplify the following:

- a)
- $(x^3y^4)^5$

$$x^{15}y^{20}$$

$$\textcircled{1} \quad \textcircled{1}$$

- b)
- $(x^8)^4 + (x^6)^2$

$$x^{32} + x^{12}$$

$$= x^{20}$$

$$\textcircled{1}$$

- c)
- $\frac{20x^7}{5x^3}$

$$\left(\frac{20x^7}{5x^3}\right)^3$$

$$\textcircled{1}$$

$$4x^4$$

$$\textcircled{1} \quad \textcircled{1}$$

$$\frac{8m^9}{n^{15}}$$

$$\textcircled{1}$$

C

- e)
- $\frac{a^3b^4 \times a^5}{a^8b}$

$$\frac{a^8b^4}{a^8b}$$

$$\textcircled{1}$$

- f)
- $\frac{9e^4 \times f^2}{7f \times 3e}$

$$\frac{9e^4f^2}{21fe}$$

$$\textcircled{1}$$

$$= b^3$$

$$\textcircled{1}$$

$$= \frac{3e^3f}{7}$$

$$\textcircled{1}$$

3. [11 Marks: 1, 2, 2, 3, 3]

- Simplify the following, expressing the answers with positive indices:

- a)
- $4x^{-3}$

$$\frac{4}{x^3}$$

$$\textcircled{1}$$

B

- b)
- $a^4b^{-4} \times a^{-6}b^8$

$$= a^{-2}b^4$$

$$\textcircled{1}$$

B

- c)
- $\frac{(x^7y^{-3})^{-2}}{x^{-3}y^{-8}}$

$$= \frac{x^{-14}y^6}{x^{-3}y^{-8}}$$

$$\textcircled{1}$$

A

$$= x^{-11}y^{14}$$

$$= \frac{y^{14}}{x^{11}}$$

$$\textcircled{1}$$

- e)
- $\left(\frac{a^{-3}b^6}{a^5b}\right)^{-4} + \frac{a^5b^{-4}}{(a^2b^{-3})^{-2}}$

$$= \frac{a^{12}b^{-24}}{a^{-20}b^{-4}} \times \frac{a^4b^6}{a^5b^{-4}}$$

$$\textcircled{1}$$

$$= \frac{C^3}{a^8b^5}$$

$$\textcircled{1}$$

$$= a^{-8}b^{-5}C^3$$

$$\textcircled{1}$$

- d)
- $\frac{a^6b^{-5}}{c^3} \times \left(\frac{a^3c^{-3}}{a^4}\right)^{-2}$

$$= \frac{a^6b^{-5}}{c^3} \times \frac{a^{-6}c^6}{a^8}$$

$$\textcircled{1}$$

A

$$= \frac{a^0b^{-5}C^6}{C^3a^8}$$

$$= \frac{C^3a^8}{a^8b^5}$$

$$\textcircled{1}$$

A

$$= \frac{a^8b^{-18}}{a^{-15}b^{-8}}$$

$$\textcircled{1}$$

$$= a^{23}b^{-10}$$

$$= \frac{a^{23}}{b^{10}}$$

$$\textcircled{1}$$

4. [10 Marks: 1, 1, 2, 2, 2, 2]
Expand and simplify the following

a) $2(x+5)$

$2x + 10$ ①

b) $5x(8-2x)$

$40x - 10x^2$ ①

c) $3a(a-2b+5c)$

$3a^2 - 6ab + 15ac$

d) $(x+3)(x-3)$

$x^2 + 3x - 3x - 9$ ①

②

$x^2 - 9$

①

e) $(x-1)^2$

$(x-1)(x-1)$

f) $2(3x-1)^2$

$2(3x-1)(3x-1)$

$x^2 - 1x - 1x + 1$ ①

$x^2 - 2x + 1$ ①

$18x^2 - 12x + 2$ ①

$2(9x^2 - 6x + 1)$

$18x^2 - 12x + 2$ ①

5. [6 Marks: 1, 1, 2, 2]
Factorise the following

a) $9x + 18$

$9(x+2)$

b) $x^2 - 5x$

$x(x-5)$

c) $x^2 - 23x - 24$

$(x+1)(x-24)$

d) $6x^2 - 6x - 36$

$6(x^2 - x - 6)$

① ①

$6(x+2)(x-3)$ ①

6. [4 Marks]

A cube has side lengths of 3 cm.
Find the surface area and volume of the cube.
Show full working out.

$TSA = (3 \times 3) \times 6$

$= 9 \times 6$ ①

$= 54 \text{ cm}^2$ ①

$V = 3 \times 3 \times 3$ ①

$= 27 \text{ cm}^3$ ①



7. [2 marks]

The irregular prism below has a base with an area of 2.5 cm^2 and is 8 cm high.
Calculate the volume of the prism.



$V = 2.5 \times 8$ ①

$= 20 \text{ cm}^3$ ①